

### 3.16 0: Statistical hypothesis testing

|    | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| O1 | Understand and apply the language of statistical hypothesis testing, developed through a binomial model: null hypothesis, alternative hypothesis, significance level, test statistic, 1-tail test, 2-tail test, critical value, critical region, acceptance region, $p$ -value; extend to correlation coefficients as measures of how close data points lie to a straight line and be able to interpret a given correlation coefficient using a given $p$ -value or critical value (calculation of correlation coefficients is excluded). |

|    | Content                                                                                                                                                                                                                                                                                                                                |
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| O2 | <p>Conduct a statistical hypothesis test for the proportion in the binomial distribution and interpret the results in context.</p> <p>Understand that a sample is being used to make an inference about the population and appreciate that the significance level is the probability of incorrectly rejecting the null hypothesis.</p> |

|    | Content                                                                                                                                                 |
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| O3 | Conduct a statistical hypothesis test for the mean of a Normal distribution with known, given or assumed variance and interpret the results in context. |